14. A family is using Liquefied petroleum gas (LPG) of weight 14.2 kg for consumption. (Full weight 29.5kg includes the empty cylinders tare weight of 15.3kg.). If it is used with constant rate then it lasts for 24 days. Then the new cylinder is replaced (i) Find the equation relating the quantity of gas in the cylinder to the days. (ii) Draw the graph for first 96days.

15. In a shopping mall there is a hall of cuboid shape with dimension $800 \times 800 \times 720$ units, which needs to be added the facility of an escalator in the path as shown by the dotted line in the figure. Find (i)the minimum total length of the escalator. (ii) the heights at which the escalator changes its direction. (iii) the slopes of the escalator at the turning points.

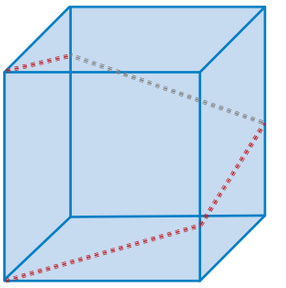


Figure 6.33

6.4 Angle between two straight lines

Two straight lines in a plane would either be parallel or coincide or intersect. Normally when two straight lines intersect, they form two angles at the point of intersection. One is an acute angle and another is an obtuse angle or equal. Both these angles would be supplements(Sum equals 180°) of each other. By definition, when we say 'angle between two straight lines' we mean the acute angle between the two lines.

$$\begin{aligned} \text{Let } y &= m_1x + c_1 \text{ and} \\ y &= m_2x + c_2 \end{aligned}$$

be the equations of two straight lines and let these two lines make angles $\theta_1$and θ_2 with x- axis.

$$\begin{aligned} \text{Then } m_1 &= \tan\theta_1 \text{ and} \\ m_2 &= \tan\theta_2 \end{aligned}$$

If ϕ (phi) is the angle between these two straight lines, then

$$\phi = \theta_2 - \theta_1 \Rightarrow \tan\phi = \tan(\theta_2 - \theta_1)$$

$$\Rightarrow \tan\phi = \frac{m_2 - m_1}{1 + m_2m_1}$$

$$\Rightarrow \phi = \tan^{-1}\frac{m_2 - m_1}{1 + m_2m_1}$$

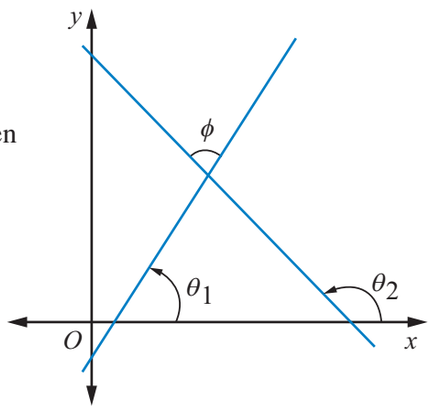


Figure 6.34

If $\dfrac{m_2 - m_1}{1 + m_2m_1}$ is positive then ϕ is the acute angle and if it is negative ϕ is the obtuse angle between the two lines. Therefore, the acute angle ϕ is $\tan^{-1}\left|\dfrac{m_2 - m_1}{1 + m_2m_1}\right|$